

# Bernard Lappay

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## WORK EXPERIENCE

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### Silent Studios

Jun. 2026 – Present

*Technical Contributor*, Mineshoku Tensei

*Remote*

- Backend, gameplay systems, and internal tooling for a live-service multiplayer RPG across multiple server instances (Java, Python, Linux).
- Diagnosed a spatial-query bottleneck in a core gameplay system and built a Python computational-geometry simplification tool that cut the affected data's complexity by **~86%**.
- Designed real-time boss encounters: multi-stage attacks driven by timing-based state machines with failsafes against race conditions.
- Closed a time-of-check/time-of-use exploit in a core gameplay interaction; patched dozens of configs with a scripted, diff-verified rollout.
- Moving world entities from ephemeral spawn/despawn to persistent, state-driven designs, probe-tested in isolation before rollout; built a resumable Python harness that audits the default permission set empirically.
- Diagnose JVM dependency and version-compatibility failures across interoperating components; keep a ledger of platform regressions.

## PROJECTS

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### Adaptive Math-Reasoning Inference Pipeline

*Python, vLLM, LoRA, RunPod H100s | CSE 151B, Spr. 2026*

- Improved a fixed 4B model (Qwen3-4B-Thinking-2507) at math via inference-time design, no retraining; ranked **#29** class-wide on Kaggle.
- Built verifier-guided majority voting (3 candidates, 2-of-3 agreement, answers normalized across multiple-choice and free-response) with adaptive retries that escalate the token budget and re-prompt with prior attempts and verifier feedback.
- Audited and repaired 7,500 distilled training examples and trained LoRA SFT/DPO adapters; deployed the adapter only as a fallback for unresolved multiple-choice questions, where it helped.
  - **Key Results:** 68/100 on the final evaluation vs. 61/100 for voting and retries alone; voting and retries lifted multiple-choice accuracy from 73.6% to 81.6% over the base model.

### Persona Chatbot & Discord Bot Platform

*Python, PyTorch, Hugging Face, llama.cpp, discord.js*

- Built a training set from **330,000** Discord messages (8 years): scrubbed personal info, merged alt accounts, split into 16,581 conversations.
- Fine-tuned Qwen3-1.7B with QLoRA on a single 12 GB GPU (64,000 examples, 36 personas), plus a DPO pass that ships only if it beats SFT on held-out chats; served locally at 4-bit.
- Two-index retrieval (a persona's own lines for voice, past conversations for facts) scoped so a persona only recalls DMs it was in; Discord bot takes turns with up to 5 users and needs no open ports.

### Handwritten Digit Recognition

*Python, NumPy, JavaScript*

- Built a 1-NN classifier from scratch over 7,500 MNIST images reaching **95.4%** test accuracy; one vectorized distance computation classifies all 1,000 test digits in  $\sim 0.25$  s, benchmarked against scikit-learn's tree search.
- Shipped a browser demo that runs the same search in JavaScript: draw a digit and see its five nearest training images.

### Self-Hosted Game Server Infrastructure

*Personal Homelab*

- Run a Minecraft network for **30+** users on a 16 GB Ubuntu mini PC; migrated it live (proxy + 3 game servers) into Docker in one day with no data lost, using a runbook and a backup-first cutover script with rollback.
- The two largest worlds start on demand and stop when empty, freeing  $\sim 7.5$  GB for a co-hosted LLM; hourly encrypted restic backups go to an append-only copy on a second machine, restore-tested.
- No admin ports exposed: web panel behind Cloudflare Tunnel with 2FA, SSH and dashboards Tailscale-only; caught Docker bypassing the host firewall and fixed it. Traced SSD resets and a crash to an NVMe firmware power-state (APST) bug, fixed with a kernel parameter.

## EDUCATION

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### University of California, San Diego

Expected Jun. 2027

*B.S., Computer Engineering*

*La Jolla, CA*

- **Relevant Coursework:** Machine Learning: Learning Algorithms; Deep Learning (CSE 151B); Probabilistic Reasoning in AI; Computer Design and Architecture (Verilog, C, MIPS); Data Networks; Data Structures (Java)

## TECHNICAL SKILLS

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- **Languages:** Python, Java (4 yrs); C, C++; JavaScript; SystemVerilog, Verilog; MIPS Assembly; MATLAB; Bash
- **ML & AI:** PyTorch; vLLM; Hugging Face (TRL, PEFT); LoRA/QLoRA fine-tuning (SFT, DPO); RAG; llama.cpp; Ollama; NumPy
- **Linux & DevOps:** Ubuntu Server; Docker; systemd; LVM; restic; Tailscale; Cloudflare Tunnel; ufw; fail2ban; iptables; tcpdump; SSH
- **Hardware & Tools:** ModelSim; Quartus; self-checking testbenches; Git; Maven; Paper API; Vim; LaTeX